

Point Cloud Object Segmentation Using PointNet and PointNet++

Group 31

Romina Bahrami¹ Nilofer Hyder² Nagham Sabbour³

Department of Electrical and Computer Engineering

Applied Deep Learning – Winter 2026

Team Contributions

Romina Bahrami – Implementation of PointNet++ Neural Network

Nilofer Hyder – PointNet++ Baseline Training, Visualization

Nagham Sabbour – Preprocessing, PointNet Baseline Training, Evaluation Metrics

GitHub Repository – <https://github.com/rominaba/3d-point-cloud-segmentation>

Data and Preprocessing

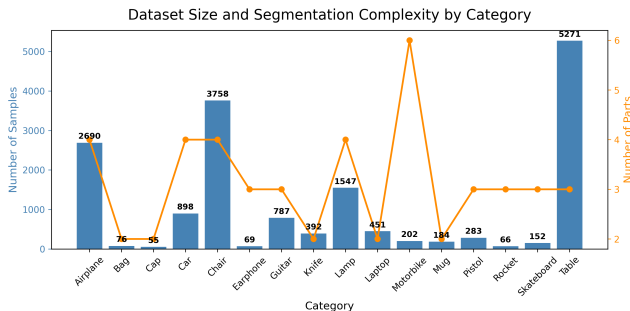
Dataset: ShapeNet Part

- 3D point cloud dataset
- 16 object categories
- Point-wise part labels for each shape
- Each class has its own set of part labels

Preprocessing

- Normalized each point cloud by centering and scaling to the unit sphere
- Randomly sampled each shape to a fixed size of 1024 points
- Used (x, y, z) coordinates without norms

Figure: The bars show total samples per category. The line shows number of part labels per category.



Model Architecture

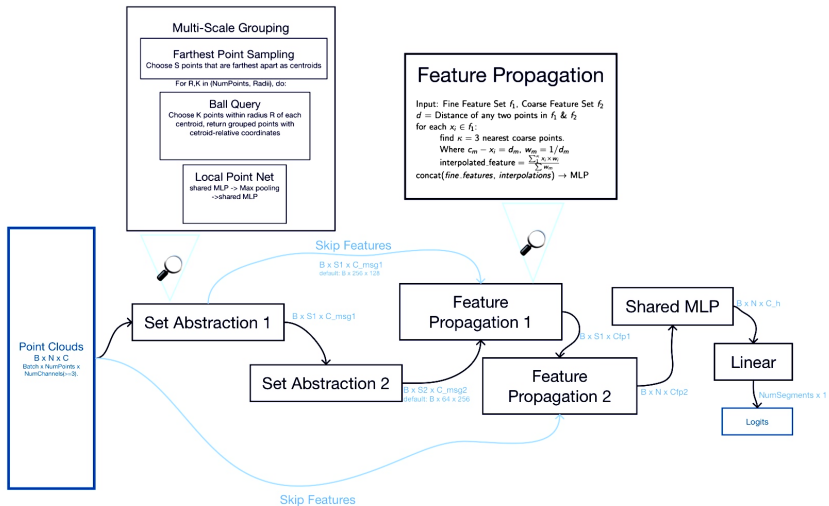


Figure: PointNet++ architecture with Multiscale Grouping for part segmentation

Evaluation Results for Part Segmentation

Architecture	Accuracy	Class Avg mIoU	Instance Avg mIoU
PointNet Baseline	0.87	0.66	0.74
PointNet++ Baseline	0.92	0.74	0.81
Our PointNet++	0.84	0.59	0.74

Table: Test results of each architecture.

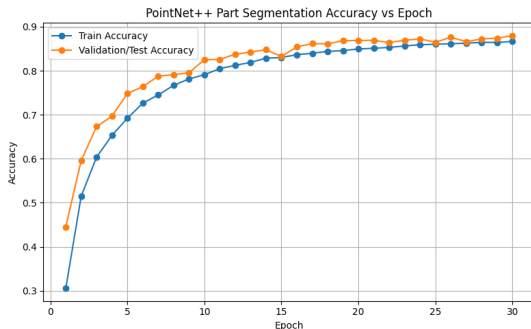
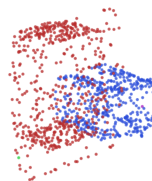


Figure: Graph of accuracy vs epochs

Knife Point Clouds



Prediction



Ground Truth

Figure: Visual comparison of part labels

Conclusion and Next Steps

Concluding Remarks

- Our PointNet++ model, while it didn't completely reach the baseline, performed fairly well, reaching within 10% of the baseline's accuracy.
- Convergence rate of our model was slower as it needed more epochs to reach a higher accuracy.

Next Steps

- Attempt to speed up the convergence rate by adding batch normalization to the feature propagation (upsampling) layer.
- Attempt to improve accuracy by applying data augmentation to undersampled categories like motorbike and rocket.